Examiner only

4. *(a)* State what is meant by a double circulatory system. [2]

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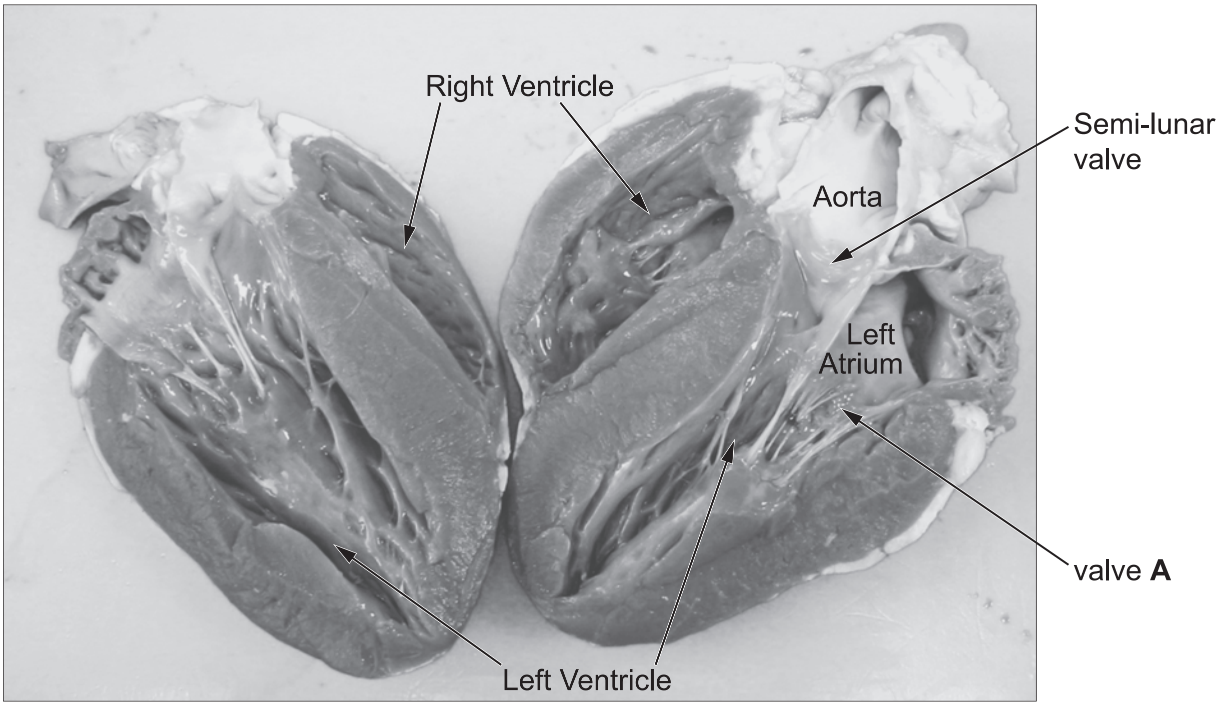
The photograph below shows a section through the heart.



(b) Use the picture above and your own knowledge to identify valve **A**. [1]

....

(c) Explain how the semi-lunar valve, shown in the photograph, helps to ensure a one-way flow of blood through the heart. [3]

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Examiner only

(d) The walls of the left ventricle are thicker than the walls of the right ventricle. Explain the significance of this difference. [2]

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In 2016, a research group at Imperial College London reported that weekly exercise seemed to increase the thickness of the walls of the ventricles. This effect could be mistaken for serious heart disease even though the individuals are healthy.

(e) Suggest an explanation for the effect of exercise on the thickness of the walls of the ventricles. [2]

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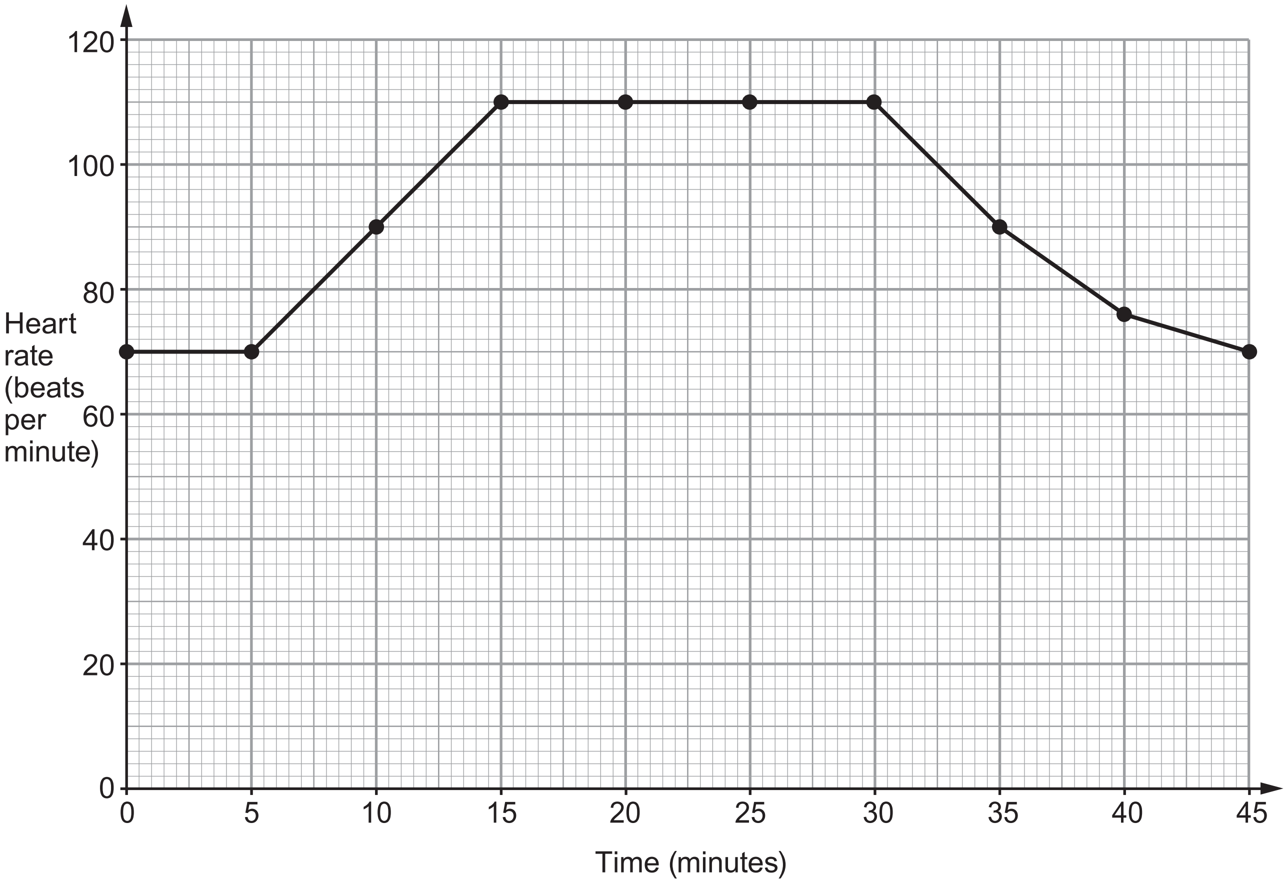
Examiner only

(f) Cardiac output is a measurement of the volume of blood pumped by a ventricle in one minute. It can be calculated as follows:

Cardiac output = volume of blood in ventricle × heart rate

The volume of blood in a ventricle of an average adult human is 70 cm^3.

The graph below shows the heart rate of an individual before, during and after a session on an exercise bike.



(i) Calculate the cardiac output at 5 minutes and 20 minutes. [1]

5 minutes = ... cm^3/min

20 minutes = ... cm^3/min

Examiner only

(ii) Calculate the percentage increase in cardiac output between 5 and 20 minutes. [2]

percentage increase in cardiac output = ... %

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